MSDM 5058 Information Science

Assignment 4 (due 25th April, 2026)

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(1) Stationary ARMA processes

A time series Y_t is governed by the following AR(2) process.

$$(1 - 1.1B + 0.18B^2)Y_t = \varepsilon_t \quad \text{for} \quad E(\varepsilon_t \varepsilon_\tau) = \begin{cases} 1 & (t = \tau) \\ 0 & (t \neq \tau) \end{cases}$$

Here B is the backward shift operator and ε_t is noise.

(a) Is it covariance-stationary? Why? (*Hint: refer to p.28 of the lecture on Time Series in Complex Systems.*)

(b) If so, calculate its autocovariance. (*Hint: refer to p.38 of the lecture on Time Series in Complex Systems.*)

Solution:

(a) Yes, it is covariance-stationary because both roots of

$$(1 - 1.1c + 0.18c^2) = (1 - 0.9c)(1 - 0.2c)$$

outside the unit circle.

(b) The autocovariance functions are

$$\gamma_0 = 1.1\gamma_1 - 0.18\gamma_2 + 1 \tag{1}$$

$$\gamma_1 = 1.1\gamma_0 - 0.18\gamma_1 \tag{2}$$

$$\gamma_s = 1.1\gamma_{s-1} - 0.18\gamma_{s-2} \tag{3}$$

Therefore,

$$\gamma_2 = 1.1\gamma_1 - 0.18\gamma_0 \tag{4}$$

From (1) and (4), we get

$$(1 - 0.18^2)\gamma_0 = 1.1(1 - 0.18)\gamma_1 + 1 \tag{5}$$

From (2) and (5), we get $\gamma_0 = 7.889$ and $\gamma_1 = 7.355$. Substituting them back into (4), we get $\gamma_2 = 6.670$. One can obtain other γ from (3).

(2) Linear regression models

You have a time series X_t which has a large number of data points. Let $\Delta X_t \equiv X_t - X_{t-1}$ be its backward difference.

(a) You have first fitted the following model

$$\Delta X_t = +0.51(\pm 0.29) \qquad - 0.11(\pm 0.04)X_{t-1}$$
$$-0.39(\pm 0.09)\Delta X_{t-1} - 0.32(\pm 0.09)\Delta X_{t-2} - 0.22(\pm 0.10)\Delta X_{t-3}$$

where the number after $\pm$ is the standard error of its preceding coefficient. Now you want to perform a unit root test.

(i) What is the t-statistic of the lagged term X_{t-1}? At what significance level can you reject the null hypothesis? (*Hint: use the ADF table provided at the end of this problem set.*)

(ii) What can you conclude if the null hypothesis is rejected?

(b) After that, you have fitted three other models

$$\Delta X_t = +0.002(\pm 0.014) - 0.31(\pm 0.10)\Delta X_{t-1} \tag{1}$$

$$\begin{aligned}\Delta X_t = &+0.021(\pm 0.158) \qquad - 0.46(\pm 0.10)\Delta X_{t-1} \\ &-0.39(\pm 0.11)\Delta X_{t-2} - 0.25(\pm 0.08)\Delta X_{t-3}\end{aligned} \tag{2}$$

$$\begin{aligned}\Delta X_t = &+1.279(\pm 0.57) \qquad - 0.51(\pm 0.10)\Delta X_{t-1} \\ &-0.44(\pm 0.11)\Delta X_{t-2} - 0.30(\pm 0.09)\Delta X_{t-3} - 0.16(\pm 0.07)Y_{t-1}\end{aligned} \tag{3}$$

where Y is an independent variable.

(i) What is the t-statistic of Y_{t-1}? At what significance level can you reject the null hypothesis for Y_{t-1}? (*Hint: use a t-distribution table.*)

(ii) You want to use models (1) to (3) and the following information to forecast X_5. Calculate their respective predicted values.

t	X_t	Y_t
1	0.8	7.7
2	4.3	7.9
3	2.9	7.7
4	1.4	7.0

Solution:

(a)

(i) The t-statistic of X_{t-1} is $0.11/0.04 = -2.75$. Since this regression model has a constant but no time trend, we should check the τ_μ statistic in the ADF table, which is -2.57 and -2.86 at the 10% and 5% significance levels respectively. Therefore, one can reject the null hypothesis at the 10% level but not the 5% level.

(ii) The rejection indicates the following equivalent facts:

– ΔX_t depends on X_{t-1};

– X_t does not have a unit root; and

– X_t is stationary.

(b)

(i) The t-statistic of Y is $-1.6/0.7 = -2.29$. As the t-statistic of two-sided events is -1.96 and -2.33 at the 5% and 2% significance levels respectively, one can reject the null hypothesis "ΔX_t does not depend on Y_t" at the 5% level but not the 2% level.

(ii) After calculating ΔX_t, substitute the values into the three regression models.

t	ΔX_t	X_t	Y_t
1	N/A	0.8	7.7
2	3.5	4.3	7.9
3	-1.4	2.9	7.7
4	-1.5	1.4	7.0

Model (1) gives

$$\Delta X_5 = 0.002 - 0.31 \cdot (-1.5) = 0.467$$
$$\Rightarrow \quad X_5 = 1.4 + 0.467 = 1.867$$

Model (2) gives

$$\Delta X_5 = 0.021 - 0.46 \cdot (-1.5) - 0.39 \cdot (-1.4) - 0.25 \cdot (3.5) = 0.382$$
$$\Rightarrow \quad X_5 = 1.4 + 0.382 = 1.782$$

Model (3) gives

$$\Delta X_5 = 1.279 - 0.51 \cdot (-1.5) - 0.44 \cdot (-1.4) - 0.30 \cdot (3.5) - 0.16 \cdot (7.0) = 0.49$$
$$\Rightarrow \quad X_5 = 1.4 + 0.49 = 1.89$$

(3) Fourier Transform of Impulses

In this question, you will learn about the Fourier Transform of impulses. Use the notations of Fourier Transform on p.58 of the lecture on Time Series in Complex Systems.

(a) Let us begin by considering a Gaussian impulse

$$f(t) = ce^{-b(t-t_0)^2}$$

where t is the time variable and (b, c, t_0) are constants. What is its Fourier transform?

(b) A Dirac delta impulse $\delta(t - t_0)$ can be approximated by using the Gaussian impulse above and setting $b = 1/(2\sigma^2)$ and $c = 1/\sqrt{2\pi\sigma^2}$.

(i) A Dirac delta function $\delta(x)$ has the properties that

$$\delta(x) = \begin{cases} 0 & (x \neq 0) \\ \infty & (x = 0) \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(x)dx = 1$$

Describe how the Gaussian impulse satisfies them as $\sigma \to 0$.

(ii) Using this approximation and the result in (a), what is the Fourier transform of a Dirac delta impulse?

(c) Next, we are going to consider some impulses that have temporal power law decay. Let us begin by considering a step function

$$f(t) = \begin{cases} 1 & (t > 0) \\ 0 & (t < 0) \end{cases}$$

If you naively perform Fourier transform on it, you will get something that is undefined. Instead, you can first perform Fourier transform on

$$f(t) = \begin{cases} e^{-\alpha t} & (t > 0) \\ 0 & (t < 0) \end{cases}$$

and then let $\alpha \to 0$ to retrieve the Fourier transform of the step function.

(i) Hence, what is the Fourier Transform of the step function? (*Hint: its discontinuity at* $t = 0$ *will give a delta function in your answer.*)

(ii) What is its power spectral distribution function? Comment on your result.

(d) Using the method shown in (c), what is the Fourier transform of the following sign function?

$$f(t) = \begin{cases} 1 & (t > 0) \\ -1 & (t < 0) \end{cases}$$

(e) Using the result in (d), what are the Fourier transforms of (i) $f(t) = 1/t$, (ii) $f(t) = |t|$, and (iii) $1/t^2$ for $-\infty \le t \le \infty$? [*Hint: for* $f(t) = 1/t$, *start from its inverse Fourier transform and redefine the parameters in the integral, i.e.* $t \to -\omega$ *and* $\omega \to t'$.]

Solution:

(a) The Fourier transform of the Gaussian impulse is

$$F(\omega) = \int_{-\infty}^{\infty} c e^{-b(t-t_0)^2} e^{-i\omega t} dt$$

For the exponent, we complete the square as follows.

$$b(t - t_0)^2 + i\omega t = b\left[t^2 - 2t_0 t + t_0{}^2 + \frac{i\omega t}{b}\right]$$

$$= b\left[\left(t - \left(t_0 - \frac{i\omega}{2b}\right)\right)^2 + \left(\frac{\omega}{2b}\right)^2 + \frac{it_0\omega}{b}\right]$$

Hence,

$$F(\omega) = c e^{-b\left[\left(\frac{\omega}{2b}\right)^2 + \frac{it_0\omega}{b}\right]} \int_{-\infty}^{\infty} e^{-b\left(t - \left(t_0 - \frac{i\omega}{2b}\right)\right)^2} dt$$

Define $t-\left(t_0-\frac{i\omega}{2b}\right)=t'$, then recall $\int_{-\infty}^{\infty} e^{-x^2}dx=\sqrt{\pi}$, so

$$F(\omega)=c\sqrt{\frac{\pi}{b}}e^{-b\left[\left(\frac{\omega}{2b}\right)^2+\frac{it_0\omega}{b}\right]}$$

(b)

(i) The Gaussian impulse becomes

$$f(t)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}(t-t_0)^2}$$

It is a Gaussian distribution with mean t_0 and variance σ^2. Since it is normalized, $\int_{-\infty}^{\infty} f(t)dt=1$ always holds. As $\sigma\to 0$, the bell shape of $f(t)$ becomes so narrow that in order to maintain the unit area, its height must approach infinity. In other words, $f(t)=\begin{cases}0 & (t\neq t_0)\\ \infty & (t=t_0)\end{cases}$.

(ii) The result in (a) becomes

$$F(\omega)=c\sqrt{\frac{\pi}{b}}e^{-b\left[\left(\frac{\omega}{2b}\right)^2+\frac{it_0\omega}{b}\right]}=c\sqrt{\frac{\pi}{b}}e^{-\frac{\omega^2}{4b}-it_0\omega}$$

$$=\frac{1}{\sqrt{2\pi\sigma^2}}\sqrt{\frac{\pi}{1/(2\sigma^2)}}e^{-\frac{\omega^2}{2/\sigma^2}-it_0\omega}=e^{-\frac{\sigma^2\omega^2}{2}}e^{-it_0\omega}$$

As $\sigma\to 0$, we get $F(\omega)=e^{-it_0\omega}$. Note that the Fourier Transform of the Dirac delta impulse has a magnitude 1 and a complex phase $t_0\omega$.

(c)

(i) The Fourier transform of the step function is

$$F(\omega)=\lim_{\alpha\to 0}\int_0^{\infty} e^{-\alpha t}e^{-i\omega t}dt$$

$$=\lim_{\alpha\to 0}\frac{1}{\alpha+i\omega}=\lim_{\alpha\to 0}\left(\frac{\alpha}{\alpha^2+\omega^2}-\frac{i\omega}{\alpha^2+\omega^2}\right)$$

To evaluate the limit, we must consider its real and imaginary parts separately. As $\alpha\to 0$,

$$\frac{\alpha}{\alpha^2+\omega^2}\to\frac{0}{0+\omega^2}\quad\text{and}\quad\frac{i\omega}{\alpha^2+\omega^2}\to\frac{i}{\omega}$$

The real part becomes indeterminate if $\omega=0$ but remains 0 otherwise. To model this behaviour, we first assume $\frac{\alpha}{\alpha^2+\omega^2}=c\delta(\omega)$ for some constant c.

On one hand, we have

$$\lim_{\alpha\to 0}\int_{-\infty}^{+\infty}\frac{\alpha \mathrm{d}\omega}{\alpha^2+\omega^2}=\lim_{\alpha\to 0}\left[\arctan\frac{\omega}{\alpha}\right]_{\omega\to-\infty}^{\omega\to+\infty}=\pi$$

On the other hand, we have

$$\lim_{\alpha\to 0}\int_{-\infty}^{+\infty}\frac{\alpha \mathrm{d}\omega}{\alpha^2+\omega^2}=\int_{-\infty}^{+\infty}c\delta(\omega)\mathrm{d}\omega=c$$

Relating the two, we get $c=\pi$. Therefore, the Fourier Transform of the step function becomes

$$F(\omega)=\pi\delta(\omega)-\frac{i}{\omega}.$$

(ii) The power spectral distribution function is $|F(\omega)|^2$, so it has a power law behavior $1/\omega^2$, which is similar to the spectrum of Brown noise as discussed in the lecture.

(d) The Fourier transform now becomes

$$F(\omega)=\lim_{\alpha\to 0}\left(\int_0^{\infty}e^{-\alpha t}e^{-i\omega t}dt-\int_{-\infty}^{0}e^{\alpha t}e^{-i\omega t}dt\right)$$
$$=\lim_{\alpha\to 0}\left(\frac{1}{\alpha+i\omega}-\frac{1}{\alpha-i\omega}\right)=\lim_{\alpha\to 0}\frac{-2i\omega}{\alpha^2+\omega^2}=\frac{-2i}{\omega}$$

(e)

(i) Instead of directly calculating the Fourier transform of $f(t)=1/t$, which involves contour integrals, we will use the duality property here. Recall that the inverse Fourier transform of a function is defined as

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)e^{i\omega t}d\omega$$

For the step function in (d), we can get it back by applying the inverse Fourier transform on $-2i/\omega$.

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{-2i}{\omega}\right)e^{i\omega t}d\omega=\begin{cases}1 & (t>0)\\ -1 & (t<0)\end{cases}$$

Redefining the parameters in the integral, i.e. $t\to-\omega$ and $\omega\to t'$, we get

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{-2i}{\omega}\right)e^{i\omega t}d\omega=\frac{-i}{\pi}\int_{-\infty}^{\infty}\left(\frac{1}{t'}\right)e^{-i\omega t'}dt'$$
$$=\begin{cases}1 & (-\omega>0)\\ -1 & (-\omega<0)\end{cases}=\begin{cases}-1 & (\omega>0)\\ 1 & (\omega<0)\end{cases}=-\operatorname{sign}(\omega)$$

Therefore, the Fourier Transform of $f(t)=1/t$ is

$$F(\omega) = \int_{-\infty}^{\infty} \left(\frac{1}{t'}\right) e^{-i\omega t'} dt' = \frac{\pi}{i}\ \mathrm{sign}(\omega) = -i\pi\ \mathrm{sign}(\omega)$$

(ii) Using the result in (d), the Fourier Transform of $f(t) = |t|$ is

$$F(\omega) = \lim_{\alpha\to 0}\left(\int_{0}^{\infty} te^{-\alpha t}e^{-i\omega t}dt + \int_{-\infty}^{0} te^{\alpha t}e^{-i\omega t}dt\right)$$

$$= \lim_{\alpha\to 0}\left(-\frac{d}{d\alpha}\int_{0}^{\infty} e^{-\alpha t}e^{-i\omega t}dt - \frac{d}{d\alpha}\int_{-\infty}^{0} e^{\alpha t}e^{-i\omega t}dt\right)$$

$$= \lim_{\alpha\to 0}\left[-\frac{d}{d\alpha}\left(\frac{1}{\alpha+i\omega} + \frac{1}{\alpha-i\omega}\right)\right]$$

$$= \lim_{\alpha\to 0}\left[-\frac{d}{d\alpha}\left(\frac{2\alpha}{\alpha^2+\omega^2}\right)\right]$$

$$= \lim_{\alpha\to 0}\left[-\frac{2(\alpha^2+\omega^2)+(2\alpha)^2}{(\alpha^2+\omega^2)^2}\right] = \frac{-2}{\omega^2}$$

(iii) Using the Fourier transform of $|t|$ and the duality property again, we get $F(\omega) = -\pi|\omega|$ for the Fourier transform of $f(t) = 1/t^2$.

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{-2}{\omega^2}\right)e^{i\omega t}d\omega = |t|$$

$$\frac{1}{\pi}\int_{-\infty}^{\infty}\left(\frac{-1}{t'^2}\right)e^{-i\omega t'}dt' = |-\omega| = |\omega|$$

$$\therefore\quad F(\omega) = \int_{-\infty}^{\infty}\left(\frac{1}{t'^2}\right)e^{-i\omega t'}dt' = -\pi|\omega|$$

Augmented Fuller-Dickey Table

Significance level	0.01	0.025	0.05	0.10
Sample Size T	The τ statistic: No Constant or Time Trend ($a_0 = a_2 = 0$)			
25	−2.65	−2.26	−1.95	−1.60
50	−2.62	−2.25	−1.95	−1.61
100	−2.60	−2.24	−1.95	−1.61
250	−2.58	−2.24	−1.95	−1.62
300	−2.58	−2.23	−1.95	−1.62
∞	−2.58	−2.23	−1.95	−1.62
	The τ_μ statistic: Constant but No Time Trend ($a_2 = 0$)			
25	−3.75	−3.33	−2.99	−2.62
50	−3.59	−3.22	−2.93	−2.60
100	−3.50	−3.17	−2.89	−2.59
250	−3.45	−3.14	−2.88	−2.58
500	−3.44	−3.13	−2.87	−2.57
∞	−3.42	−3.12	−2.86	−2.57
	The τ_τ statistic: Constant + Time Trend			
25	−4.38	−3.95	−3.60	−3.24
50	−4.15	−3.80	−3.50	−3.18
100	−4.05	−3.73	−3.45	−3.15
250	−3.99	−3.69	−3.43	−3.13
500	−3.97	−3.67	−3.42	−3.13
∞	−3.96	−3.67	−3.41	−3.12

Source: The table is reproduced from Fuller (1996).